

SEPM Top 10 Direct Exam Answers (From Notes)

1. Software Project Management

Software Project Management is the process of planning, organizing, directing, and controlling software development activities so that the project is completed within time, cost, and quality limits. Main activities include project planning, scheduling, resource allocation, risk management, and monitoring progress.

2. Software Project Life Cycle

Project Life Cycle means the sequence of phases through which a software project passes from beginning to end: Initiation, Planning, Execution, Monitoring and Control, and Closing.

3. Waterfall Model

Waterfall Model is a linear software development model in which each phase is completed before moving to the next phase. Phases: Requirement Analysis, System Design, Coding, Testing, Deployment, Maintenance. Advantages: simple and easy. Disadvantages: difficult to change requirements, testing done late.

4. Agile Development

Agile is an iterative software development approach focusing on continuous delivery and customer feedback. Features include small iterations, customer involvement, fast delivery, and flexibility.

5. RAD Model

RAD (Rapid Application Development) is a fast software development model based on prototyping and quick user feedback. Phases: Requirements Planning, User Design, Construction, Cutover.

6. Feasibility Study

Feasibility Study checks whether a software project is practical before development starts. Types: Technical, Economic, Legal, Operational, Schedule (TELOS).

7. Requirement Elicitation

Requirement Elicitation is the process of collecting requirements from users and stakeholders. Techniques: Interview, Questionnaire, Brainstorming, FAST.

8. Functional and Non-Functional Requirements

Functional Requirements define what the system does, such as login and report generation. Non-Functional Requirements define quality attributes like security and performance.

9. Data Flow Diagram (DFD)

DFD shows movement of data inside the system. Components: Process, Data Flow, Data Store, External Entity.

10. Software Project Manager

Software Project Manager manages software project from planning to completion. Responsibilities include planning, cost estimation, risk handling, and team coordination.